

AI Multi-Disease Prediction

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PROBLEM STATEMENT

Healthcare systems worldwide struggle with timely and accurate disease identification. Delayed diagnosis increases patient risk and healthcare costs.

GOAL: Build an AI model that classifies one of 10 disease conditions from patient vitals and symptoms using supervised machine learning with a Random Forest Classifier.

Key Challenge: Multi-class classification with overlapping symptom patterns across 10 disease categories.

DISEASE CLASSES (10 Categories)

- * 1. Flu - Fever + Cough + Fatigue + High Temp
- * 2. Common Cold - Cough + Runny Nose + Sore Throat
- * 3. Allergy - Skin Rash + Runny Nose + Headache
- * 4. Migraine - Headache + Fatigue (no fever)
- * 5. Food Poisoning - Nausea + Vomiting + Fatigue
- * 6. Skin Infection - Skin Rash + Fever
- * 7. Asthma - Wheezing + Cough
- * 8. Hypertension Risk- High BP + Age > 40
- * 9. Diabetes Risk - High Glucose + Age > 35
- * 10. Healthy - No significant condition

DATASET

Records : 5,000 (synthetic)
Features: 15 input variables

Numeric:
age (5-84)
temperature_f (93.8-103F)
systolic_bp (95-184 mmHg)
glucose (70-239 mg/dL)

Categorical:
sex (male / female)

METHODOLOGY

Step 1: Dataset Generation
Rule-based synthetic data
for 10 disease classes (n=5000)

Step 2: EDA
Distribution plots,
Correlation heatmap,
Class balance analysis

Step 3: Preprocessing
One-Hot Encoding (sex)
80/20 Train-Test Split

MODEL RESULTS

~90%
Accuracy

RF
Model

4,000
Training Records

TECH STACK

- * Python 3.11 - Core language
- * pandas & NumPy - Data manipulation
- * scikit-learn - ML pipeline & Random Forest
- * matplotlib - Visualisations
- * seaborn - Statistical plots
- * Streamlit - Interactive web application
- * Jupyter Notebook - Analysis & reporting
- * GitHub - Version control & hosting

KEY FINDINGS

- * High-temperature + cough + fatigue strongly predicts Flu
- * Wheezing is the dominant feature for Asthma detection
- * Glucose > 200 at age > 35 reliably flags Diabetes Risk
- * Systolic BP > 160 at age > 40 indicates Hypertension Risk
- * Skin rash combined with fever differentiates Skin Infection
- * Random Forest handles overlapping symptom patterns well
- * Model achieves ~90% accuracy on unseen test data
- * All 10 classes correctly predicted by the Streamlit app

ML PIPELINE

Data Generation

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EDA & Visualization

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Pre- processing

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Model Training

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Evaluation

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Streamlit Deployment

LIVE STREAMLIT APPLICATION

<https://1roahan3-droiroahn-ai-multi-disease-prediction-app-wpqa56.streamlit.app/>

GitHub Repository

https://github.com/1roahan3-droiroahn/ai_multi_disease_prediction